ESSnet Big Data

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Work Package 4

AIS Data

Deliverable 4.1

Creating a database with AIS data for official statistics: possibilities and pitfalls

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1. Introduction

ESSnet BIG DATA is part of the Big Data Action Plan and Roadmap 1.06. It was agreed to integrate it into the ESS Vision 2020 portfolio. The related business case Big Data received the support of the ESSC at its meeting on 20 June 2015 in Luxemburg. The overall objective of the project is to prepare European Statistical System (ESS) for the integration of big data sources into the production of official statistics. A consortium of 22 partners, consisting of 20 National Statistical Institutes and 2 Statistical Authorities has been formed in September 2015 to meet the objectives of the project. According to the Framework Partnership Agreement (FPA) between the consortium and Eurostat, the project runs between February 2016 and May 2018. The consortium has subdivided its work into 8 work packages (WP) each WP dealing with one pilot and a concrete output.

Aim of WP 4 is to investigate whether real-time measurement data of ship positions (measured by the AIS system) can be used (1) to improve the quality and internal comparability of existing statistics and (2) for new statistical products relevant for the ESS.
WP4 is subdivided into two phases: SGA-1 and SGA-2. SGA-1 runs from February 2016 to July 2017 and has three deliverables planned:

Deliverable 1: report on creating a database with AIS data for official statistics: possibilities and pitfalls (this report)

Deliverable 2: report about deriving harbour entries and linking data from port authorities with AIS data. This report also gives more information about the data quality of AIS data.

Deliverable 3: report about sea traffic analyses using AIS data

SGA-2 runs from April 2017 to May 2018. SGA-2 concentrates on calculating emissions, investigating possibilities for new statistical output based on AIS data and describes future perspectives of using AIS data for official statistics. Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for WP 8 (Methodology) of SGA-2. There is no direct link between WP4 and other WP's in the ESSnet Big Data programme. See Annex 1 for a more detailed description of WP4.

There are five National Statistical Institutes participating in WP4: the national statistical institutes of the Netherlands (Work package leader), Denmark, Greece, Norway and Poland.
Chapter 2 will discuss the maritime statistics as they are currently collected by Eurostat, the wishes from the different parties and how this is connected to the products we wish to make in this work package.
Chapter 3 gives a short introduction into AIS data. Chapter 4 describes which decisions we had to make about obtaining AIS data at the European level and which alternative sources of European AIS data we investigated. Chapter 5 describes the decisions we had to make concerning tools and environment. Chapter 6 describes how we pre-processed the data and upload the data to the UNECE. Finally, in chapter 7 we present our conclusions about the first 6 months of WP4.
The first appendix states in more detail the tasks of WP4 and the deliverables. The second appendix includes the technical commands for handling the data.

2. Maritime Statistics

The current maritime statistics are described in directive 2009/42/EC of the European Parliament and the Council. It consists of ten tables, and seven of them are about carriage of goods. One table is about the transportation of passengers and two are about vessel traffic. More details about the directive can be found at the Eurostat website for Maritime statistics legislation and methodology.

In 2011 the European Commission for Mobility and Transport released a White Paper titled "**Roadmap to a Single European Transport Area – Towards a competitive and resource efficient transport system"** in which they set out their goals for the coming years. For maritime transport it states: "The environmental record of shipping can and must be improved by both technology and better fuels and operations: overall, the EU CO_2 emissions from maritime transport should be cut by 40 % (if feasible 50 %) by 2050 compared to 2005 levels". This can't be measured if there are no statistics about emissions of maritime transport.

We hope that, with AIS, we can estimate the vessel traffic as defined in the maritime statistics directive. If the tests are positive, this would yield a single data source that all member states could use to provide these statistics. Member states could exchange software and code to process the data. This would not only lower the administrative burden but also the burden on the member states to make the statistics. Uniform data quality would also be guaranteed because of the single data source and the same process tools and quality. This would be the first time that international statistics would be made from a single source. Also we hope to make an estimate of the emissions from maritime transport to monitor the environmental record of shipping for the European Commission for Mobility and Transport.

3. AIS data

Maritime traffic has increased exponentially over the last decades. This requires different solutions to ensure safety at sea. Radar was developed in the 1940s, but this technology could only detect ships that were in line of sight. Another technology that was launched in the 20th century was the Global Positioning System (GPS). It was developed for US defence to ensure that US troops would always be able to find their bearings anywhere in the world. Based on this technology, the Automatic Identification System (AIS) was able to broadcast their location and status information over a radio channel, making it possible to detect other ships wherever they were, as long as a radio signal could be sent and received. This AIS data is collected at several places around the world.
Within Europe EMSA, Kystverket, Hellenic Coastguard, Dirkzwager and Marine Traffic all collect data. The number of applications for this data is enormous, and some apps are already built on the basis of it. For official statistics, AIS data could be used to create marine traffic statistics, which could be interesting for estimating emissions or identifying locations at sea with a critical amount of traffic. Furthermore, the AIS data could be used as a backbone for the maritime transport statistics. Based on AIS we know how a ship, that entered one harbour also entered other harbours. In this document, we report on the

activities and the decisions we made concerning the activities in the ESSnet WP4 regarding collecting and pre-processing the AIS data.

4. Obtaining AIS data on European level

The first hurdles to take were deciding the level for which we should collect the data and obtaining the data at that level. There were three levels to choose from:

- National level for each country collaborating in WP4
- European level: all waters within the perimeter of the European countries
- World level: a data set covering the whole world.

In table I, the three levels of data are scored against some criteria which we viewed as important, i.e.: potential use for European statistics, potential use for national statistics, price of the data, data size.

Table I: scoring the different levels of data.

	National	European	World Wide
Potential for EU level statistics	-	++	+++
Potential use for national statistics	++	+++	++++
Data size	+++	++	+
Data price	+++	+	-

National data is not very useful for European analysis. The only advantages are pricing (most of the participating countries already have national data) and size (not all that big).
European data is much more suited for European and national statistics. Since European data contains information about the routes vessels take within Europe, the data gives information of routes, but also about harbours of loading and unloading within Europe. Hence, this data could also have potential extra value for the national level.
With world-wide data one would be able to locate the harbours of loading and unloading around the world. Furthermore, this could be interesting for calculating the environmental footprint of certain goods transported over sea. However, this data is gathered by satellites and as a consequence it is expensive. At this moment we don't have an exact estimate, but we don't think we can get this data for less than € 100,000 a year. Because of our tight budget we didn't explore this option in more detail.
Based on the fact that we would like to make some statistics about European waters, plus the extra information for national statistics we would be able to gather based on the European data, it was decided to obtain European data. Several sources were identified to purchase the data from, i.e.:

- EMSA
- Kystverket
- Hellenic Coastguard
- Dirkzwager
- Marine Traffic (.com)
- Joint Research Center (JRC)

We will describe these sources in the next sections.

4.1 European Maritime Safety Authority (EMSA)

The European Maritime Safety Agency (EMSA) is one of EU's decentralised agencies. Based in Lisbon, the Agency provides technical assistance and support to the European Commission and Member States in developing and implementing EU legislation on maritime safety, pollution by ships and maritime security. It has also been given operational tasks in the field of oil pollution response, vessel monitoring and in the long range identification and tracking of vessels.

EMSA is an EU agency under the authority of DG MOVE. DG MOVE is the transport policy Directorate General (DG) of the European Commission. DG MOVE is responsible for the legal framework and governing of the EU maritime databases, while EMSA is responsible for the technical operation of this databases. Access to AIS data is therefore granted by DG MOVE and the High Level Steering Group (HLSG) of SafeSeaNet (SSN, representatives for the maritime authorities in the Member States), while EMSA will provide the data if told to do so by DG MOVE.

Eurostat has already worked with AIS data from EMSA. They provided AIS data for the year 2011. Currently Eurostat is discussing with EMSA and DG MOVE whether to get an extraction of AIS data for 2014 for the purpose of exploring if a model using AIS data to complement the Port notifications sent by vessels can improve the current EU vessel port call statistics. This request has been approved by the HLSG. Eurostat is currently negotiating the terms of the data access agreement (which has to be signed between Eurostat and EMSA before the data is handed over). Eurostat has asked to have a clause included allowing the 2014 AIS data extraction to be shared with WP4 of the ESSnet Big Data project. If this is accepted, we can test the 2014 AIS data within the framework of the ESSnet Big Data project. Unfortunately, EMSA seems of the opinion that sharing the AIS data with WP4 of the ESSnet Big Data Programme is not within the mandate for this extraction approved by the HLSG. But this is still being discussed.

If the ESS decides, after WP4, that it will need continued provision of AIS data from the SSN Central system (EMSA) or the SSN National systems (in each Member State), this kind of data access will have to be requested from the HLSG (for access to data from the SSN Central system) or the maritime authorities in each Member State (for access to data from the respective SSN National system). In case data is needed from the SSN Central system (hosted by EMSA), a formal request describing the planned project will have to be sent to DG MOVE by either Eurostat or the ESS.

Statistics Netherlands has also tried to get AIS data from EMSA, but it seems better to try and obtain AIS data from EMSA via Eurostat. So we decided not to further investigate the possibilities of Statistic Netherlands obtaining AIS data from EMSA. We will wait what the results are of Eurostat request for the 2014 AIS data to also be made available for this WP.

4.2 Kystverket

The main focus of Kystverket (The Norwegian Coastal Administration) is the sea around Norway, but they also have data for other European countries. Statistics of traffic around Norway is available through www.havbase.no and they are working on policy/solution for making some of the underlying data public available. The commercial company, DNV-GL, makes calculations for Kystverket. Kystverket has agreed to share the national AIS data freely with Statistics Norway as input for statistical products.

When it comes to data for other European countries, Kystverket consulted their legal advisors. They found out that sharing such data goes beyond their authorization. They

believe that "each NSI must go to the AIS owner in their own country or buy AIS data commercially from a market player such as DNV-GL, Marine Traffic or similar."

4.3 Hellenic Coast Guard (HCG)

The Hellenic Cost Guard (HCG) is a public authority that operates under the Ministry of Maritime Affairs and Insular Policy in Greece. The HCG plays a crucial role as the competent administration and the policymaker in the field of maritime industry and shipping. The HCG plans, develops, deploys and operates Systems and IT infrastructure for the effective maritime surveillance in order to fulfil its mission. One of the basic elements of this infrastructure is the coastal AIS base station network.

In reply, to our request for European AIS data, they informed us that they can provide us only with national historical AIS data since the second half of 2015. In their opinion, we should address our request for European AIS data at European Maritime Safety Agency (EMSA) in accordance with their established procedures for data acquisition.

4.4 Dirkzwager

Dirkzwager is a private company which provides the maritime industry with data. They have AIS data for the European waters based on land based stations and satellite data. They provide this data to shipping companies to follow their ships across the world. We arranged for WP4 that Dirkzwager would deliver 6 months (8 October 2015- 12 April 2016) of data at the very special price of 2,000 euros. These contain AIS data for the European waters based on land based stations. Satellite data is not included. Normally one month of these data from Dirkzwager would cost 7,500 euros, so 6 months of data would be 42,000 euros. The data consist of the raw AIS data with a timestamp of the date and time they received the AIS data from the live stream they use.

4.5 Marine Traffic

Marine Traffic is one of the most popular ship tracking services. Marine Traffic collects real-time vessel position data and uses them to create applications. Marine Traffic is a global pioneer in AIS vessel tracking. Around to 800 million vessel positions and 18 million vessels and port related events are recorded monthly. A request for European AIS data had been sent to them. A potential collaboration with Marine Traffic is under investigation.

4.6 Joint Research Centre (JRC)

The role of the Joint Research Centre (JRC), is the in-house science service of the European Commission. The JRC draws on over 50 years of scientific work experience and continually builds its expertise based on its seven scientific institutes located in Belgium (Brussels and Geel), Germany, Italy, the Netherlands and Spain. While most of their scientific work serves the policy Directorates General of the European Commission, they address key social challenges while stimulating innovation and developing new methods, tools and standards. They share know-how with the Member States, the scientific community and international partners. The JRC collaborates with over a thousand organisations worldwide whose scientists have access to many JRC facilities through various collaboration agreements.

We had a Webex meeting with JRC to investigate what they already did with AIS data and what the possibilities are of sharing their AIS data with our WP. JRC has a lot of experience with AIS data. Unfortunately, we cannot use their data because of legal issues, but we can use their experience in saving, cleaning and analysing AIS data.

4.7 Final Choice: use AIS data from Dirkzwager in this WP

The result of this investigation on possible sources for European AIS data is that we know for sure we cannot obtain European data from Kystverket, Hellenic Coastguard and JRC. We are still investigating the possibilities of getting European AIS data by Marine traffic, but this seems to be a very expensive alternative exceeding our budget. We prefer using AIS data from EMSA because they provide the EMSA data for free, but the process to obtain the AIS data by EMSA takes a long time. Dirkzwager could provide us with 6 months of European AIS data on a short period for a very good price. That is why we decided to use this Dirkzwager data within our WP. If the EMSA data is available during SGA-1 or SGA-2 we will also investigate this data in WP4.

During the negotiations we wanted the raw AIS data. This data is internationally standardized and can be decoded in the same way. The difference is the area they cover. National data only covers the national coast line, whereas international data covers several coastal areas. Satellite data provides information on vessels on the open sea.

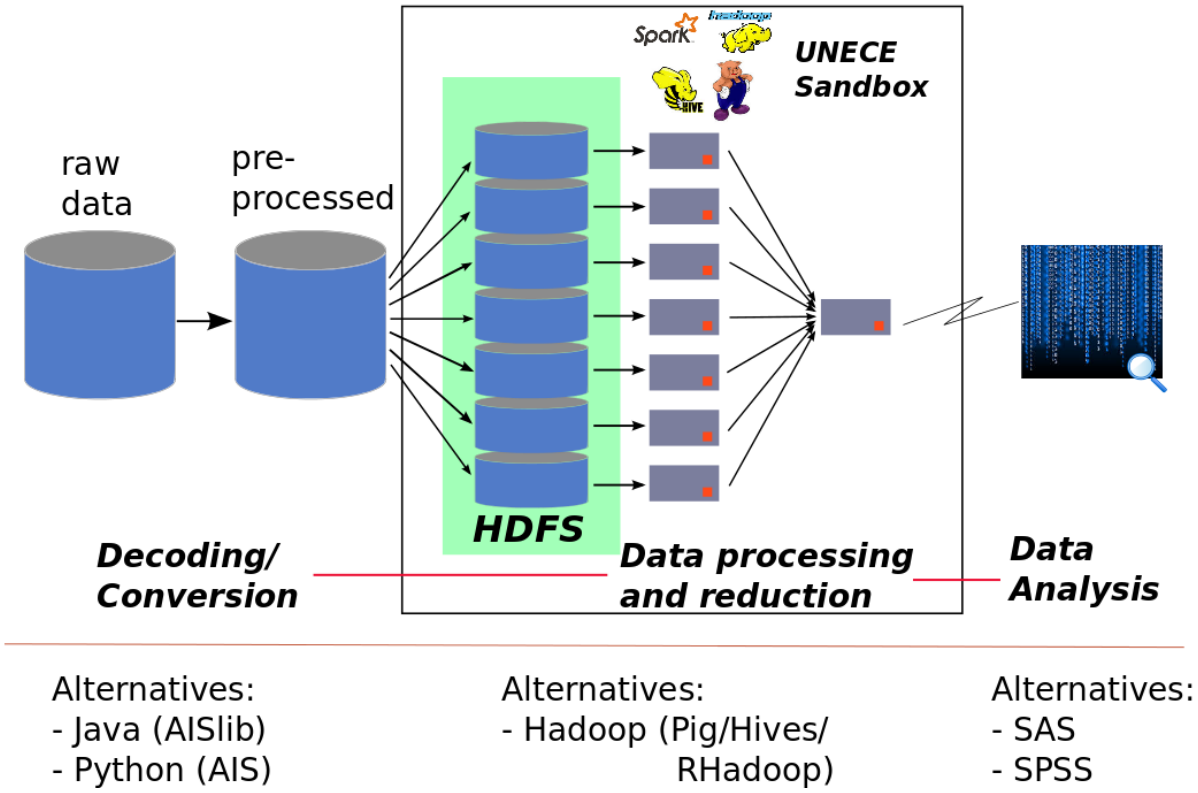
5. Decisions concerning tools and environment

Some decisions were made concerning the tools to be used and the location where the data is going to be processed and stored. As Figure 1 shows, processing and storing the data consists of several steps. First the data is pre-processed, and the pre-processed data is moved to its data store.

There were two possibilities to decode the data:

- At statistics Netherlands, python was used in combination with the AIS package
- At statistics Denmark, Java was used in combination with the libAIS library.

Figure 1: pre-processing, processing and storing the AIS data



For legal reasons we choose to keep the data in the Netherlands and process the data in python. After processing, the data had to be moved to a data storage. Since the aim of the project is to get hands-on experience in Big Data, it was decided that the data should be processed using the Hadoop Stack. Since the data should be accessible for all participating countries, it was decided to store the data at the UNECE sandbox in HDFS. From the HDFS, the data can be accessed using the tools that are available in the Hadoop stack, which are:

- Pig
- Hive
- RHadoop
- Spark

Pig is a big data tool that makes use of the language Pig Latin, developed by Yahoo. It is an easy accessible language for using Hadoop and expressing data flows. It supports the loading and processing of input data with many operators (arithmetic, relational) that transform the input data to the desired output. **Hive** is very similar to SQL language called as HiveQl (Hive Query Language) developed by Facebook, to create simple SQL queries and analysis of data in Hadoop. **RHadoop** is an integration of the statistical language R in Hadoop. The integration is convenient and makes it possible to do analysis on Big Data. Finally, **Spark** is the new kid on the block. This makes it possible to perform much more complex processing

on data which is stored in the HDFS. It can be used in combination with the programming languages Scala, Python, R and Java. However, it is advised to use it with Python or maybe Scala. Furthermore, within Spark it is possible to write SQL queries using SparkSQL. Because Spark is the most flexible tool, most of our work will be done in Spark. However, the use of the other tools is not prohibited. Spark is accessible via SSH (SSH is a Secure Shell - an encrypted command line interface for remote computers), Pig and Hive are accessible via HUE (the web interface for Hadoop), which also incorporates a file browser. RHadoop is available over a web version of RStudio.
For the final step it is decided that resulting aggregates will be downloaded from the UNECE Sandbox using HUE. Researchers of the different NSIs will be able to analyse the data in their tools of choice, i.e. SPSS, SAS, R, or even import the data into a local database and use that for analysis. It also will be investigated if the data can be accessed by using JDBC or ODBC. However, analyses can also be done by using the web version of RStudio with RHadoop. The advantage is that it is unnecessary to download the data from the sandbox.

6. Pre-processing and uploading the data

As Figure 1 shows, the data is processed in several stages. The received AIS messages from Dirkzwager are in NMEA format, which is a text encoded binary format (see the 6th field in Figure 2), which has to be decoded. The dataset was about 400 GB in size.

Figure 2. Example of raw NMEA encoded AIS data

```
!AIVDM,1,1,,A,13aEOK?P00PD2wVMdLDRhgvL289?,0*26
!AIVDM,1,1,,B,16S`2cPP00a3UF6EKT@2:?vOr0S2,0*00
!AIVDM,2,1,9,B,53nFBv01SJ<thHp6220H4heHTf2222222222221?50:454o<`9QSlUDp,0*0
9
!AIVDM,2,2,9,B,888888888888880,2*2E

description of the NMEA format, in order:

!AIVDM - The NMEA message type: !AIVDM (received data from other vessels)
and !AIVDO (your own vessels information)
2 - Total numer of sentences needed to transfer the message (1 to 9)
2 - Sentence number (1 to 9)
9 - Sequential message ID (0 to 9)
B - AIS Channel (A or B)
888888888888880 - The Encoded AIS Data
2*- End of Data
2E - NMEA Checksum
```

It is important to note that some messages are distributed over several lines. Line 3 and 4 in figure 2 encode one message. You can see this because the first element after the AIVDM tag is a 2, denoting that the message is split into 2 parts. The element after that denotes which part of the message it is, so we can see that the third line is the first, and the 4th line is the second part. The two strings in the 6th column have to be concatenated before sending to the decoder.

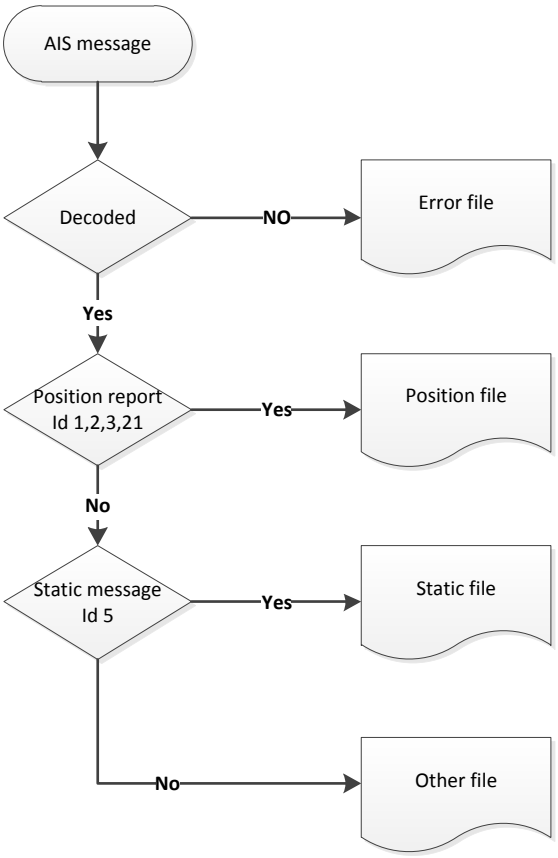
As mentioned earlier, the messages were decoded using the Python AIS module. This library was already used in the Netherlands. However, the program crashed several times when

decoding the European dataset. It took some time to figure out on which part of the dataset it actually crashed, since the error occurred 10 times on the in total 4.5 billion cases.

AIS consist of several types of records. We divided the messages into four file types. Some are about the position of the vessel (where the message id is 1, 2, 3 or 21), whereas others are about voyage related issues (where the message id is 5). The third was all other AIS messages we could decode, but the message type was not in the position report or voyage related. At this point we don't see any information in this data, but perhaps in the future there will be some data we can use. The fourth is data we couldn't decode. The records were written to different sets of files. One set concerned the position of the vessel and the others concerned the voyage of the vessel.

The set of files with positions of the vessel was the biggest. The position files have the following fields. The timestamps are technically not part of the VDM messages, but are transmitted as separate messages in the AIS stream.

Figure 3: dividing AIS messages into four types of files



After the data was decoded, all files were zipped to save space. The sets of files with positions and files with voyage related data were uploaded to the UNECE sandbox by a secure copy (scp). It took approximately 7 hours to copy the data.

Ships are linked between the two datasets (which were uploaded to the UNECE sandbox) by the Maritime Mobile Service Identity (mmsi) and approximately by the timestamp. The filenames are built up as follows: <year><month><day><hour><minute>.csv.gz. The two datasets are distinguished by placing them in the folders Locations and Messages respectively. The locations data comprises 144 GB of compressed data (about 200 GB uncompressed) and the messages comprised about 5 GB of compressed data (about 7 GB uncompressed). After the data was decoded, it was uploaded to the UNECE sandbox by a secure copy (scp). It took approximately 7 hours to copy the data.

After uploading, the data was copied to the Hadoop Distributed File System (HDFS) and available for use. Sometimes, the ssh-interface of the UNECE sandbox freezes. It is therefore advisable to use the screen-command, not only for uploading the data, but also for running a bigger job in Spark.

The data will be made accessible within the HDFS filesystem at the location /datasets/AIS/Messages and /datasets/AIS/Locations. Legal issues made us decide to create an AIS group on the Sandbox, so only the usernames from the members of WP4 can access the Dirkzwager data.

7. Conclusion

At the start of the project we had to make some decisions about which AIS data to obtain and where to process it. We decided to obtain the European data from a company called Dirkzwager and use the UNECE sandbox to process the data. The advantage of the sandbox is that it is open for all NSIs within WP4. In the future it can be open for all NSIs within the UNECE. After the data was obtained, it was transformed and uploaded to the UNECE sandbox and stored in the Hadoop Filesystem (HDFS). The data will be made available in the HDFS at the location /datasets/AIS/ where two directories are available, i.e. Locations and Messages. We were able to run some first queries on the data, as seen in the previous paragraph, which brings us to the conclusion that we have created a good starting point for the next phase of the project.

Annex A: WP4 in more detail

Work package number	4	Start date: End date:				1.2.2016 31.7.2017	
Title	**AIS Data**						
Partner/co-beneficiary	**NL** **120**	DK 60	EL 40	NO 67	PL 40		

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the AIS system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS.

Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations.

New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS data at European level is that the source data are generic word wide and data can be obtained at the European level.

Challenges ahead with this dataset are: obtaining the data at the European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. It is especially important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for the envisaged WP 8 of SGA-2, in case SGA-2 will be realised. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

Task 1 – Data access.

AIS data are available for national territories and the entire European territory. For example, AIS data in the Netherlands are provided by Rijkswaterstaat (a government agency which is part of the Ministry of Infrastructure and the Environment). It is expected that similar agencies in other countries have the AIS data for their national territories.

At the European level, a dataset of AIS data is available at the European Maritime Safety Agency (EMSA). The advantage of using one AIS dataset for the entire European territory is (a) a better comparison of international traffic between the countries and (b) more synergy as all participating countries work on the

same dataset. A disadvantage is that these data are stored by private companies and handling fees have to be paid. Aim of this task is to decide how European data could be used for this project, to investigate the possibilities of acquiring data from EMSA (to be coordinated with Eurostat) and, if European data are too costly or too hard to obtain, how national datasets can be obtained.

This task will involve:

- Exploration of the possibilities to collect the data at a European (or worldwide) level.

Participants: NL, DK, EL, NO, PL

Task 2 – Data handling

Aim of this task is to process and store the data in such a way that they can be used for consistent multiple outputs, like

- linking AIS data with data from port authorities
- traffic analyses
- Inference of journeys from AIS data.

Key elements of this task are:

- which programming language and environment should be used for transformation
- where will the data be processed (in each NSI, by NSIs at European level, by data holders)
- how can we create an environment which is easily accessible for all partners

Participants: NL, NO, PL, and possibly DK (if national sources are used)

Task 3 – Methodology and Techniques

Develop traffic statistics: Linking with data from port authorities.

AIS data may be linked to data from port authorities. Added value of linking AIS data to data from port authorities is that the same reference population (= ship number) is used in all harbours. As the journeys and harbour calls of ships can be derived from AIS, this linking provides the ESS information about the origin/destination of the cargo, too.

Aims of this task are:

- to build a reference frame of ships in European water (based on AIS data)
- to find out how data from port authorities can be linked to AIS data
- to check whether information improves the quality of current statistical outputs and provides more information about the origin/destination of the cargo

Participants: NL, NO, PL

Traffic analyses

The number of ships during a certain time interval at certain coordinates (like inland waterways or at certain points at sea) can be calculated by AIS data. This possibility will be explored because this information could be interesting for traffic analyses as well as economic analyses.

Aims of this task are:

- calculate the number of ships at certain coordinates
- visualise the results to analyse variations in time

Participants: NL, NO, PL, EL

Estimate emissions (envisaged under SGA-2)

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is 1) to infer journeys from AIS data 2) visualise the results, 3) combine these journeys with a model to calculate emissions and 4) estimate the impact of carrying out these calculations at the European level on the quality of emissions calculations.

Task 4 – Future perspectives (envisaged under SGA-2)

Aim of this task is to summarise the project results and perform a qualitative cost-benefit analysis of using AIS data for official statistics. These analyses should include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure.

Deliverables (SGA-1 only):

4.1	Report on creating a database with AIS data for official statistics: possibilities and pitfalls	month 6
4.2	Report about deriving harbour calls and linking data from port authorities with AIS-data	month 12
4.3	Report about sea traffic analyses using AIS data	month 18

Milestones (SGA-1 only):

4.4	Progress and technical report of first internal WP meeting	month 4
4.5	Progress and technical report of second internal WP meeting	month 9

Annex B: Technical specifications

B1: Record types

The set of position files include message with id's 1, 2, 3 and 21. It consists of the following variables:

- mmsi
- lon
- lat
- accuracy
- speed
- course
- rotation
- status
- timestamp

The set of files with voyage related data had message id 5 and consist of the following variables:

- mmsi
- imo
- name
- callsign
- destionation
- draught
- dim_a
- dim_b
- dim_c
- dim_d
- fix_type
- type and cargo
- timestamp

B2: Uploading to the sandbox

Code used for uploading the data files on the sandbox:

```
hdfs dfs -put Messages/* AIS/Messages/
hdfs dfs -put Locations/* AIS/Locations/
```

Using the screen function for when the sandbox freezes:
Before we start the job, we type:

```
screen
```

After which we start the job at hand, e.g.:

```
spark-submit xxx.py
```

We hit ctrl-a and x to exit the screen and let the job run. If we want to reconnect to the screen we type:

```
screen -r
```

After which we can see if the job finished.

B3: Data availability

In this chapter, it will be demonstrated how the data will be made available from Spark. It is assumed that the Spark context is available under the name sc as is standard in the Spark shell. After we started the spark shell, we type the statement below to get all data available in Spark.

```
val rawdata = c.textFile("hdfs://namenode.ib.sandbox.ichec.ie:8020/
datasets/AIS/Locations/*.csv.gz");
```

The variable 'data' is a so called resilient distributed dataset (RDD). If one only wants to analyse the data of a certain period, the wildcard *.csv.gz can be changed. For instance, if one wants to analyse the 2015 data, one can do:

```
var rawdata = c.textFile("hdfs://namenode.ib.sandbox.ichec.ie:8020/
datasets/AIS/Locations/20151231*.csv.gz”);
```

After we loaded the data, we have to split the lines to be able to access the individual fields. This is done by the following line:

```
var splitdata = rawdata.map(_.split(“,”));
```

As a next step we have to get rid of the headers in the data. Normally we start the header with a hash sign to indicate it as the header line. This makes filtering the headers out much easier. However, this was not done in the case of the AIS data. For that reason we have to scan for the fieldname of the first field:

```
var data = splitdata.filter(x => x(0)!=”mmsi”);
```

Now, the RDD data only contains data without headers and can we access the data. For instance, a record count for that particular day we selected:

```
data.count();
```

gives us: res3: Long = 24358293 as an answer. So, at December 31 2015, 24,358,293 records were collected. One can exit spark with the exit command.